

Lensing-Dynamics Mass Discrepancy as a Redshift-Dependent Signature of Informational Gravity

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<https://github.com/hmahaffeyges/IAM-Validation>

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Abstract

The Informational Actualization Model (IAM) predicts that gravitational lensing and dynamical mass estimates for galaxy clusters should disagree by a redshift-dependent factor $M_{\text{lens}}/M_{\text{dyn}} = 1/\mu(z)$, where $\mu(a) = H_{\Lambda}^2(a)/[H_{\Lambda}^2(a) + \beta_m E(a)]$ is the gravitational coupling parameter with $\beta_m = \Omega_m/2 = 0.1575$ and $E(a) = \exp(1 - 1/a)$. At $z = 0$ this predicts $M_{\text{lens}}/M_{\text{dyn}} = 1/0.864 = 1.157$, a 15.7% excess of lensing mass over dynamical mass. This discrepancy vanishes at high redshift as $\mu \rightarrow 1$, following a specific functional form determined entirely by IAM's activation function with no free parameters. The redshift evolution uniquely distinguishes IAM from hydrostatic bias (constant with z), MOND (acceleration-dependent), and standard modified gravity theories (which generically predict $\mu > 1$, producing the opposite sign). We compute the predicted $M_{\text{lens}}/M_{\text{dyn}}(z)$ curve, compare to existing cluster mass discrepancy measurements from Planck SZ, the Canadian Cluster Comparison Project, and Weighing the Giants, and identify the observational tests that will confirm or challenge this signature with forthcoming Euclid, Rubin, and eROSITA data.

1 Introduction

Galaxy cluster mass estimates obtained through gravitational lensing consistently exceed those obtained through dynamical methods — galaxy velocity dispersions and X-ray hydrostatic equilibrium — by 10–30% Mahdavi2013,Israel2014, Hoekstra2015,vonDerLinden2014,Smith2016. This discrepancy is conventionally attributed to hydrostatic bias: the assumption that the intracluster medium is in perfect hydrostatic equilibrium, neglecting non-thermal pressure support from bulk motions, turbulence, and cosmic rays. The standard parametrization is:

$$M_{\text{true}} = \frac{M_{X\text{-ray}}}{1 - b} \quad (1)$$

with observational estimates ranging from $b \approx 0.1$ to $b \approx 0.4$ depending on methodology Nagai2007,Rasia2012,Biffi2016.

The hydrostatic bias explanation has several unsatisfying features. The required bias is larger than predicted by hydrodynamic simulations ($b \approx 0.1$ – 0.15 from Lau2009,Nelson2014). The bias shows no clear dependence on cluster properties that would be expected if it were purely a gas physics effect. And the tension between Planck SZ cluster counts and Planck CMB cosmology — which requires $b \approx 0.4$, far above simulation predictions — suggests the explanation may not be purely astrophysical PlanckClusters2016.

In this paper, we show that the Informational Actualization Model Mahaffey2026a,Mahaffey2026b naturally produces a lensing-dynamics mass discrepancy with the correct magnitude and a distinctive redshift dependence that no other proposed explanation shares. The prediction follows directly from IAM’s unique signature: $\mu < 1$ (weakened gravitational coupling for matter) with $\Sigma = 1$ (unmodified light deflection). Full Boltzmann validation through 15 converged Planck 2018 MCMC chains confirms $\Delta\chi^2 = +0.54$ relative to Λ CDM with $\sigma_8 = 0.800$ Mahaffey2026a.

2 The IAM Prediction

2.1 The μ – Σ framework

In the standard modified gravity parametrization Pogosian2016,Amendola2008, deviations from General Relativity are captured by two functions of scale factor:

$$\nabla^2\Psi = -4\pi G a^2 \mu(a) \rho\delta \quad (\text{Poisson equation, dynamics}) \quad (2)$$

$$\nabla^2(\Psi + \Phi) = -8\pi G a^2 \Sigma(a) \rho\delta \quad (\text{lensing potential}) \quad (3)$$

where Ψ is the Newtonian potential governing matter dynamics and $\Phi + \Psi$ is the lensing potential governing photon deflection. In GR, $\mu = \Sigma = 1$.

2.2 IAM values

IAM predicts Mahaffey2026a,Mahaffey2026b:

$$\mu(a) = \frac{H_\Lambda^2(a)}{H_\Lambda^2(a) + \beta_m E(a)}, \quad \Sigma(a) = 1 \quad (\text{exact, all redshifts}) \quad (4)$$

where $H_\Lambda^2(a) = \Omega_m a^{-3} + \Omega_\Lambda$, $\beta_m = \Omega_m/2 = 0.1575$ (virial theorem, derived), and $E(a) = \exp(1 - 1/a)$ (activation function, derived from horizon thermodynamics). No parameters are fitted.

The physical origin of $\Sigma = 1$ is fundamental: photons travel on null geodesics ($d\tau = 0$) and do not experience gravitational decoherence. The informational entropy contribution that modifies gravity for matter is invisible to light. Lensing sees the full, unmodified gravitational potential.

2.3 Mass ratio prediction

A cluster’s dynamical mass is inferred from matter responding to the Ψ potential:

$$M_{\text{dyn}} \propto \Psi \propto \mu \times M_{\text{true}} \quad (5)$$

A cluster’s lensing mass is inferred from photons responding to the $(\Psi + \Phi)/2$ potential:

$$M_{\text{lens}} \propto \frac{\Psi + \Phi}{2} \propto \Sigma \times M_{\text{true}} = M_{\text{true}} \quad (6)$$

Therefore:

$$\boxed{\frac{M_{\text{lens}}}{M_{\text{dyn}}} = \frac{\Sigma}{\mu} = \frac{1}{\mu(z)}} \quad (7)$$

This is a zero-parameter prediction. The entire $M_{\text{lens}}/M_{\text{dyn}}(z)$ curve is determined by $\beta_m = \Omega_m/2$ and $E(a) = \exp(1 - 1/a)$.

Table 1: Predicted lensing-to-dynamical mass ratio $M_{\text{lens}}/M_{\text{dyn}} = 1/\mu(z)$ as a function of redshift. Computed from Equation 4 with $\beta_m = 0.1575$ and Planck 2018 parameters. The 15.7% excess at $z = 0$ declines monotonically to zero at high redshift.

Redshift z	$\mu(z)$	$M_{\text{lens}}/M_{\text{dyn}}$	Excess (%)
0.0	0.864	1.157	15.7
0.1	0.886	1.129	12.9
0.2	0.905	1.105	10.5
0.3	0.922	1.085	8.5
0.5	0.948	1.055	5.5
0.7	0.966	1.035	3.5
1.0	0.982	1.018	1.8
1.5	0.994	1.006	0.6
2.0	0.998	1.002	0.2
3.0	1.000	1.000	0.0

2.4 Explicit functional form

Substituting Equation 4 into Equation 7:

$$\frac{M_{\text{lens}}}{M_{\text{dyn}}}(a) = 1 + \frac{\beta_m E(a)}{H_{\Lambda}^2(a)} = 1 + \frac{\beta_m \exp(1 - 1/a)}{\Omega_m a^{-3} + \Omega_{\Lambda}} \quad (8)$$

This is the observable prediction: a monotonically declining function of a (increasing function of z), fully determined by Planck 2018 parameters and the virial theorem, with no fitted parameters.

3 Comparison with Alternative Explanations

3.1 Hydrostatic bias

The standard hydrostatic bias b produces a constant mass ratio $M_{\text{lens}}/M_{\text{dyn}} = 1/(1 - b) \approx 1.2$ for $b = 0.17$, independent of redshift by construction. Some redshift dependence might arise from cluster evolution (relaxation state), but this would correlate with dynamical state indicators and would not follow the specific functional form of Equation 8.

Distinguishing test: Plot $M_{\text{lens}}/M_{\text{dyn}}$ vs. z . Hydrostatic bias predicts a flat line or scatter with no systematic trend. IAM predicts a specific declining curve.

3.2 MOND and MONDian theories

Modified Newtonian Dynamics and its relativistic extensions predict enhanced gravitational effects at low accelerations ($a < a_0 \approx 1.2 \times 10^{-10} \text{ m s}^{-2}$). The mass discrepancy depends on the acceleration profile of each individual cluster, not on cosmological redshift.

Distinguishing test: At fixed cluster mass and morphology, MOND predicts $M_{\text{lens}}/M_{\text{dyn}}$ depends on position within the cluster. IAM predicts it depends only on the cluster’s observed redshift, independent of internal properties.

3.3 Standard modified gravity ($f(R)$, nDGP)

Most modified gravity theories predict $\mu > 1$ — gravity is stronger than GR, not weaker. This produces $M_{\text{dyn}} > M_{\text{lens}}$, the opposite of what is observed. The few models that allow $\mu < 1$ typically also predict $\Sigma \neq 1$, which conflicts with the unmodified CMB lensing power spectrum measured by Planck Planck2020lens.

IAM is unique in the current literature in predicting $\mu < 1$ with $\Sigma = 1$ exactly, arising from a physical mechanism (decoherence asymmetry between timelike and null worldlines) rather than parameter tuning.

3.4 Summary of distinguishing signatures

Table 2: Predicted lensing-dynamics signatures for competing models. IAM is the only model predicting $\mu < 1$ with $\Sigma = 1$ exactly and a specific, parameter-free redshift dependence.

Model	$M_{\text{lens}}/M_{\text{dyn}}$ $z = 0$	at	z -dependence	μ vs. Σ
GR + hydrostatic bias	~ 1.2 (fitted b)		Flat or scatter	$\mu = \Sigma = 1$
MOND	Varies with a/a_0		Acceleration-dependent	$\mu > 1, \Sigma > 1$
$f(R)$ gravity	< 1 ($M_{\text{dyn}} > M_{\text{lens}}$)		Scale-dependent	$\mu > 1, \Sigma > 1$
nDGP gravity	< 1		z -dependent	$\mu > 1, \Sigma \geq 1$
IAM	1.157		Eq. 8, zero params	$\mu < 1, \Sigma = 1$

4 Existing Observational Evidence

4.1 Planck SZ cluster counts

The Planck SZ cluster count analysis PlanckClusters2016 found a significant tension between the number of detected clusters and the prediction from Planck CMB cosmology. Resolution requires either $S_8 \approx 0.78$ (well below the CMB value of 0.83) or a hydrostatic bias $b \approx 0.4$ (far above simulation predictions of $b \approx 0.1$ – 0.15). IAM resolves this naturally: $\mu < 1$ suppresses σ_8 from 0.811 to 0.800, reducing the predicted cluster count

without requiring extreme hydrostatic bias. The residual discrepancy is consistent with a moderate $b \approx 0.1\text{--}0.15$, in line with simulations.

4.2 Canadian Cluster Comparison Project (CCCP)

Hoekstra et al. Hoekstra2015 compared weak lensing masses to X-ray hydrostatic masses for 50 massive clusters at $0.15 < z < 0.55$, finding $M_{\text{lens}}/M_X = 1.20 \pm 0.12$ on average. The IAM prediction at the median redshift $z \approx 0.3$ is $M_{\text{lens}}/M_{\text{dyn}} = 1.085$, within the measurement uncertainty. CCCP did not bin by redshift with sufficient resolution to test the z -dependence.

4.3 Weighing the Giants (WtG)

Von der Linden et al. vonDerLinden2014 performed weak lensing calibration of Planck cluster masses, finding $M_{\text{lens}}/M_{\text{Planck}} = 1.31 \pm 0.11$ for the full sample. The IAM prediction at the sample median $z \approx 0.25$ is 1.095, below the WtG central value. However, the WtG comparison includes Planck mass estimation systematics beyond hydrostatic bias, making a direct comparison model-dependent.

4.4 The discrepancy pattern

Across multiple analyses, the consistent finding is that lensing masses exceed dynamical or X-ray masses by 10–30%. The magnitude is consistent with IAM’s $z = 0$ prediction of 15.7%. The scatter is consistent with varying cluster redshifts, dynamical states, and measurement systematics overlaid on a systematic IAM signal. No existing analysis has performed a high-precision test of the redshift evolution in the mass discrepancy using a uniform methodology across a wide redshift baseline. This is the critical test.

5 Predictions for Forthcoming Surveys

5.1 Euclid

The Euclid mission Euclid2022 will provide weak lensing mass estimates for $\sim 10^5$ galaxy clusters out to $z \approx 2$. Combined with spectroscopic velocity dispersions, this enables a direct $M_{\text{lens}}/M_{\text{dyn}}$ comparison in fine redshift bins. The IAM predictions from Equation 8:

Table 3: IAM predictions for $M_{\text{lens}}/M_{\text{dyn}}$ in Euclid redshift bins, computed from Equation 8 with no free parameters. The declining trend from 10.5% to $< 1\%$ over $0.2 < z < 2.0$ is the falsifiable signature.

Redshift bin	$M_{\text{lens}}/M_{\text{dyn}}$ (IAM)	Excess (%)
$z = 0.2$	1.105	10.5
$z = 0.5$	1.055	5.5
$z = 1.0$	1.018	1.8
$z = 1.5$	1.006	0.6
$z = 2.0$	1.002	0.2

If the discrepancy is flat (hydrostatic bias) or acceleration-dependent (MOND), IAM is challenged. If it follows the declining curve of Equation 8, that is a direct measurement of $\mu(z)$ from cluster data.

5.2 Rubin Observatory LSST

The Legacy Survey of Space and Time LSST2009 will provide photometric cluster catalogs and weak lensing mass maps. The statistical power of $\sim 200,000$ clusters will enable percent-level mass ratio measurements in narrow redshift bins, directly testing the shape of the predicted curve.

5.3 eROSITA

The eROSITA X-ray survey Merloni2012 provides X-ray hydrostatic mass estimates for $\sim 100,000$ clusters. Cross-matching with Euclid and Rubin weak lensing gives a uniform lensing-vs-X-ray comparison sample spanning $0 < z < 1.5$.

5.4 Combined test design

The optimal test compares $M_{\text{lens}}/M_{\text{dyn}}$ in at least five redshift bins spanning $0.1 < z < 2.0$ using a uniform mass estimation methodology: (1) select clusters with both weak lensing and spectroscopic velocity dispersion measurements; (2) apply consistent NFW mass models to both; (3) compute $M_{\text{lens}}/M_{\text{dyn}}$ in each bin with bootstrap errors; (4) fit to three models: constant (hydrostatic bias), IAM (Equation 8, zero free parameters), and power law $M_{\text{lens}}/M_{\text{dyn}} = 1 + A(1+z)^n$ (two free parameters); and (5) compare via BIC/AIC. The key discriminant is the functional form of the z -dependence, not just its existence.

6 Connection to IAM Cosmology

The lensing-dynamics mass discrepancy is one manifestation of IAM’s fundamental prediction: the sector split between matter and photon observables. The same $\mu < 1$, $\Sigma = 1$ signature produces the Hubble tension ($H_0^{\text{matter}} = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ vs. $H_0^{\text{photon}} = 67.16 \text{ km s}^{-1} \text{ Mpc}^{-1}$), the σ_8 suppression from 0.811 to 0.800, and the $f\sigma_8(z)$ suppression at $z < 1$ Mahaffey2026a. All follow from $\beta_m = 0.1575$ and $E(a) = \exp(1 - 1/a)$ without additional fitting. The lensing-dynamics test is particularly powerful because it is measurable with current and near-future data, does not depend on CMB modeling, and has a distinctive functional form in redshift.

7 Discussion

7.1 Systematic concerns

The primary systematic in any lensing-dynamics comparison is the calibration of both mass estimates. Weak lensing masses depend on source redshift distributions, shape measurement systematics, and projection effects. Dynamical masses depend on assumptions

about velocity anisotropy and dynamical equilibrium. X-ray masses depend on temperature calibration and the hydrostatic assumption.

IAM’s prediction applies to the *ratio* $M_{\text{lens}}/M_{\text{dyn}}$, which cancels many systematics that affect absolute mass estimates. The redshift dependence provides additional discriminating power: systematics that evolve with z would need to conspire to mimic the specific functional form of Equation 8, which is physically unmotivated for any known systematic.

7.2 Prior work on μ – Σ from cluster masses

Several authors have constrained the μ/Σ ratio from cluster mass comparisons Terukina2014, Wilcox2015. Pizzuti et al. Pizzuti2017 used cluster dynamics and lensing to constrain modified gravity models. These analyses have generally focused on constant μ , Σ values rather than redshift-dependent functions. IAM’s specific prediction — $\mu(a)$ declining monotonically from 1 at high z to 0.864 at $z = 0$, with $\Sigma = 1$ exactly — has not been tested in this context.

7.3 Falsifiability

IAM’s lensing-dynamics prediction is explicitly falsifiable. If $M_{\text{lens}}/M_{\text{dyn}}$ shows no z -dependence, the hydrostatic bias interpretation is preferred. If it increases with z , standard modified gravity is preferred. If $M_{\text{lens}}/M_{\text{dyn}}$ at $z = 0$ differs from 1.157 by more than 3σ with sufficient sample size, the normalization is challenged. If the z -dependence follows Equation 8, that constitutes a direct measurement of $\mu(z)$ from cluster data. Whatever the surveys find, the comparison to Equation 8 is unambiguous.

8 Conclusion

The Informational Actualization Model predicts a lensing-dynamics mass discrepancy $M_{\text{lens}}/M_{\text{dyn}} = 1/\mu(z)$ of 15.7% at $z = 0$, declining monotonically to zero at high redshift following the specific, parameter-free functional form of Equation 8. This prediction is consistent with the 10–30% lensing excess observed in multiple cluster surveys. The redshift evolution provides a unique discriminant against hydrostatic bias (flat in z), MOND (acceleration-dependent), and standard modified gravity (opposite sign). Upcoming surveys from Euclid, Rubin, and eROSITA will provide the statistical power to test the predicted redshift evolution at percent-level precision, offering a direct measurement of $\mu(z)$ from cluster data.

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